
EDUCATION

- **Institute of Engineering, Tribhuvan University** Kathmandu, Nepal
Bachelor of Engineering in Computer Engineering 2023

RESEARCH EXPERIENCE

- **ATOM Research Group** Remote
Research Collaborator (Supervisor: Dr. Lancelot Da Costa, Prof. Bernhard Schölkopf) Oct 2025 – Present
 - **World Model Induction:** Contributed to Pinductor, a framework for inducing executable POMDP models from raw observation-action trajectories using LLM priors, eliminating the need for privileged hidden-state supervision.
 - **Ablation Studies:** Led extensive ablation studies across MiniGrid environments to evaluate model robustness under stochastic dynamics and verify the hypothesis around LLM-driven hypothesis generation.
- **Manifold Research Group** Remote, USA
Open Science Research Fellow July 2025 – Jan 2026
 - **MultiNet v1.0:** Designed and implemented a systematic benchmark to evaluate architectural limitations of Vision-Language-Action (VLA) models across 11 tasks, covering vision-language understanding, robotic control, multi-agent planning/coordination, and tool use. Investigated core incompatibilities between multimodal reasoning and action-generation capabilities in current foundation model architectures.
 - **MultiNet v2.0:** Defined task suites and capability dimensions for assessing the generality and robustness of VLA and generalist models across a broad range of domains. Developing dynamic and evolving benchmarks.
- **Summer of Open AI Research (Organized by Eluther AI)** Remote
Researcher (Supervisor: Dr. Catherine Arnett) Aug 2025 – Nov 2025
 - **Empirical Analysis of Tokenization-Performance Relationships:** Evaluated multiple tokenizers and multiple models across varying vocabulary sizes, analyzing effects on model perplexity, efficiency, and cross-lingual transfer capabilities.
- **Institute for Research and Innovation in Intelligent Systems (IRIIS)** Kathmandu, Nepal
Founder & Researcher July 2025 – Present
 - **Multilingual Planning, Reasoning and Human-AI Coordination Understanding and Benchmark:** Developing planning & reasoning datasets and trying to understand model capabilities to coordinate, plan & solve tasks in multi-agent (human or ai) environments, and developing strong benchmarks along the way.
 - **Educational Material Development:** Creating intuition-driven lecture content for probability theory and CUDA programming, employing visual and interactive approaches inspired by 3Blue1Brown.
- **Information and Language Processing Research Lab (ILPRL)** Kavre, Nepal
Research Collaborators (Supervisor: Prof Bal Krishna Bal) Aug 2024 – June 2025
 - **Nepali Corpus Construction for Foundation Models:** Assembled and processed 27.5GB Nepali text corpus, contributing to low-resource language data and addressing current challenges.
 - **Monolingual Language Model Pre-training:** Trained language models (BERT, RoBERTa & GPT-2) for Nepali language. Implemented instruction tuning to evaluate efficiency in low-resource settings.
 - **NLUE: Comprehensive NLU Evaluation:** Contributed 12 tasks to the Nepali General Language Understanding Evaluation benchmark, enabling systematic assessment of Nepali NLP systems.
- **Modulo Research** Cambridge, UK
Research Intern (Supervisor: Dr. Gabriel Recchia) June 2024 – Aug 2024
 - **SAFE:** Conducted evaluation on the factual consistency of long-form text generation using the Search Augmented Factuality Evaluation (SAFE) method, enabling oversight through search-based verification of individual claims
 - **Evaluation:** Executed experiments to evaluate performance of SAFE and LLMs, analyzed results using metrics such as F1@k, and compared SAFE outputs with human annotations.

- **AI Research Center, ACEM**

Kathmandu, Nepal

Research Assistant (Supervisor: Asst. Prof Bipun Man Pati)

Oct 2023 – Feb 2024

- **Research Grant Coordination:** Performed literature reviews and drafted grant proposals targeting Kathmandu Metro, ICIMOD SANDEE, and NAST.
- **Student Project Supervision:** Guided undergraduate teams through complete ML project lifecycles: automatic speech recognition for Nepali and glaucoma detection and segmentation from retinal imagery.

RESEARCH

- Six*, V., Panse*, F., Fajeau*, M., Costa*, L. D., **Sharma†, M.**, Amayuelas†, A., Xiao, T. Z., Hyland, D., Hennig, P., & Schölkopf, B. **Learning POMDP World Models from Observations with Language-Model Priors.** (Under Review @ NeurIPS 2026 Conference).
- Lopardo, A., Salim, L. F., Srivastava, S., Jiang, K., **Sharma, M.**, & Arnett, C. **No Single Tokenizer Feature Reliably Predicts Downstream Language Model Performance.** (Under Review @ ARR).
- *Guruprasad, P., *Chowdhury, S., *Sikka, H., ***Sharma, M.**, *Lu, H., *Rivera, S., *Ren, H., Khurana, A., & Wang, Y. **Do Multimodal Foundation Models Truly Generalize? Exposing Failure Modes Across Perception, Reasoning, and Action** . MMFM Workshop, CVPR 2026.
- Chang, T. A., Arnett, C., **Sharma, M.**, & 334 others. **Global PIQA: Evaluating Commonsense Reasoning Across 100+ Languages and Cultures.** (Under Review @ NeurIPS 2026 Evaluations and Datasets Track).
- Recchia, G., Mangat, C., Nyachhyon, J., **Sharma, M.**, Canvan, C., Epstein-Gross, D., & Abdulbari, M. **Confirmation bias: A challenge for scalable oversight.** AI Alignment Track, AAAI 26.
- *Nyachhyon, J., ***Sharma, M.**, Thapa, P., & Bal, B. K. **Consolidating and Developing Benchmarking Datasets for the Nepali Natural Language Understanding Tasks.** IJCNLP-AAACL 2025
- *Thapa, P., *Nyachhyon, J., ***Sharma, M.**, & Bal, B. K. **Development of pre-trained transformer-based models for the Nepali language.** CHiPSAL Workshop, COLING 2025.

VOLUNTEER EXPERIENCE

- **AAAI-26 AI Alignment Track (2025-26):** Peer reviewer for submissions on technical alignment, scalable oversight, and mechanistic interpretability.
- **MRL 2025 Shared Task:** Created PIQA-style physical commonsense reasoning dataset for Nepali language.
- **Cohere for AI Global MMLU (2024):** Verified translation accuracy and cultural appropriateness of Nepali language benchmark items.

PROFESSIONAL EXPERIENCE

- **Lamida, Inc.**

Remote, USA

AI Engineer

Aug 2025 – Present

- **Digital Twin Models:** Developed and designed protocols for mental health specialist models using behavioral cloning.
- **Video Analysis:** Developing video analysis pipeline for criminal activity detection, classification and traffic violation.
- **Adversarial Robustness:** Designed mechanisms against prompt injection attacks in Better-therapy subnet.

- **Keeper Dating, Inc.**

Remote, USA

Machine Learning Engineer

Feb 2025 – July 2025

- **Ranking Optimization:** Enhanced ranking architecture for large-scale user matching through learned strategies.
- **Spam Detection:** Developed classification system for fraudulent profiles using behavior and content analysis.
- **Model Distillation:** Distilled GPT-4o and DeepSeek-R1 for efficient reranking with reduced latency.

- **Virtly IT & Business Solutions Sarl (ICEBRKR)**

Machine Learning Engineer

Kathmandu, Nepal

Mar 2024 – Aug 2024

- **Dialogue Summarization:** Fine-tuned models for conversational summarization optimized for mobile devices.
- **User Matching:** Designed collaborative filtering algorithm with temporal dynamics for compatibility scoring.

REFERENCES

- **Prof. Dr. Bal Krishna Bal**

Associate Dean, Kathmandu University

Professor, Department of Computer Science and Engineering, School of Engineering

Email: bal@ku.edu.np

- **Dr. Lancelot Da Costa**

Postdoc, ELLIS Institute Tübingen

Email: lancelot.dacosta@gmail.com

- **Dr. Gabriel Recchia**

Director, Modulo Research Ltd

Email: gabe@moduloresearch.com